Austria or Australia?

Can CLIP ViT-B/32 tell two countries apart from images alone, or does it only know the postcards?

Zero-shot · Country211 · text side + image side

github.com/cristianknebel/AI-and-Visualization



Austria



Australia

The Visionaires: Cristian · Alexander · Shamekh · Eiman

Two letters and half a world apart. Can a model see the difference?

- Austria ↔ Australia: famously confused by name, but visually?
- CLIP ViT-B/32 (OpenAI), pre-trained, used zero-shot: no training on our data.
- Country211: real geotagged photos from 211 countries, including both of ours.
- Question: does CLIP use place evidence or stereotypical scene cues?

3 label sets

Only the words change: Austria / Australia, Österreich / Australien, or a mixed pair.

2 studies

The text side (Pair 1) and the image side (Pair 2), same model.

Two pairs, One model!

Pair 1 · text side · AI-aided

Cristian Knebel & Alexander Strasser

- Do the words in the prompt change the verdict?
- 3 label sets: English / German / mixed
- Prediction Explorer UI + Language Shift scatter

Pair 2 · image side · no LLMs

Shamekh Al-Suwi & Eiman Mohamed

- Does the image content decide, and where do errors live?
- Embedding space: t-SNE map, k-means clusters, density
- Scene probes + nearest-neighbour checks

Same model, different question

Prompt wording vs image content: the two halves of CLIP.

Different data slices

500 photos (train + test) for Pair 1; 600 photos (all splits) for Pair 2. Both zero-shot.

PAIR 1 METHOD

Same photos, three sets of words

EN · Austria / Australia | **DE · Österreich / Australien** | **Mixed · Österreich / Australia**

- 500 photos: 250 Austria + 250 Australia (Country211 train + test), streamed once and cached.
- CLIP scores every photo under all three label sets, plus the top-5 of all 211 countries.
- Prediction Explorer: photo on the left, one bar chart per label set, ✓ / ✗ verdict, close-call margin, “Only mistakes” browser.
- Language Shift: one dot per photo, Austria score in English (x) vs German (y). Reads the same cache; no model needed.

500 photos

250 AT + 250 AU · images and predictions cached in the repo

3 label sets

Only the words change; the 500 images never do

PAIR 1 FINDINGS

The label language moves the verdict

86.6% → 76.8%

accuracy, EN labels → DE labels (mixed: 82.8%)

482 / 500

photos get a lower Austria score when the names are German

93 flips

EN says Austria, DE says Australia. 71 are Austrian photos (22 AU).

Not symmetric

The drop hits Austrian photos hardest: CLIP leans on the English token.

139 / 500

Photos wrong under at least one of the three label sets.

Take-away

CLIP isn't reading "Austria". It's reading the token.

482/500 and 93 flips are printed in the committed figure (language_shift_scatter.png). Accuracy values are derived from Pair 1's cached predictions.json, recomputable with Language shift.py.

PAIR 2 METHOD

Not *how often*: where and why

600 photos

150 train + 50 valid + 100 test per country. Nothing trained; every split unseen.

17 %

zero-shot error with “a photo taken in …” prompts (102 of 600)

5 prompt pairs

The same image re-scored under 5 wordings to test prompt fragility

Embed all photos (512-d), then map the space: t-SNE 2-D map + k-means, with k picked by silhouette → k = 9 (sil 0.05).

Image-side check: does a 10-nearest-neighbour country vote separate AT and AU on its own?

Scene probes: 14 CLIP text prompts (“a photo of snow”, …) estimate scene content; explicitly unreliable.

Zoom in: sub-cluster the worst cluster and compare its halves.

PAIR 2 FINDINGS

Errors live where photos carry no place information

220 photos

4 place-specific clusters (Vienna streets, AU coast, alpine snow, field hockey): error ≤ 7%

314 photos

52% of the data. 4 content-poor clusters (night skies, conferences, windows/graffiti, concerts) hold ≈ 80% of all errors

69 : 33

Austrian vs Australian photos misclassified: the bias points towards Australia

Sparser → worse

Dense single-scene clusters (0.75) err 0–2%; sparse mixed ones (0.63–0.64) err 24–35%.

Stereotypes, not places

Snow & buildings → Austria; wildlife & night → Australia. 20 of 27 shown errors: AT photo → “Australia”.

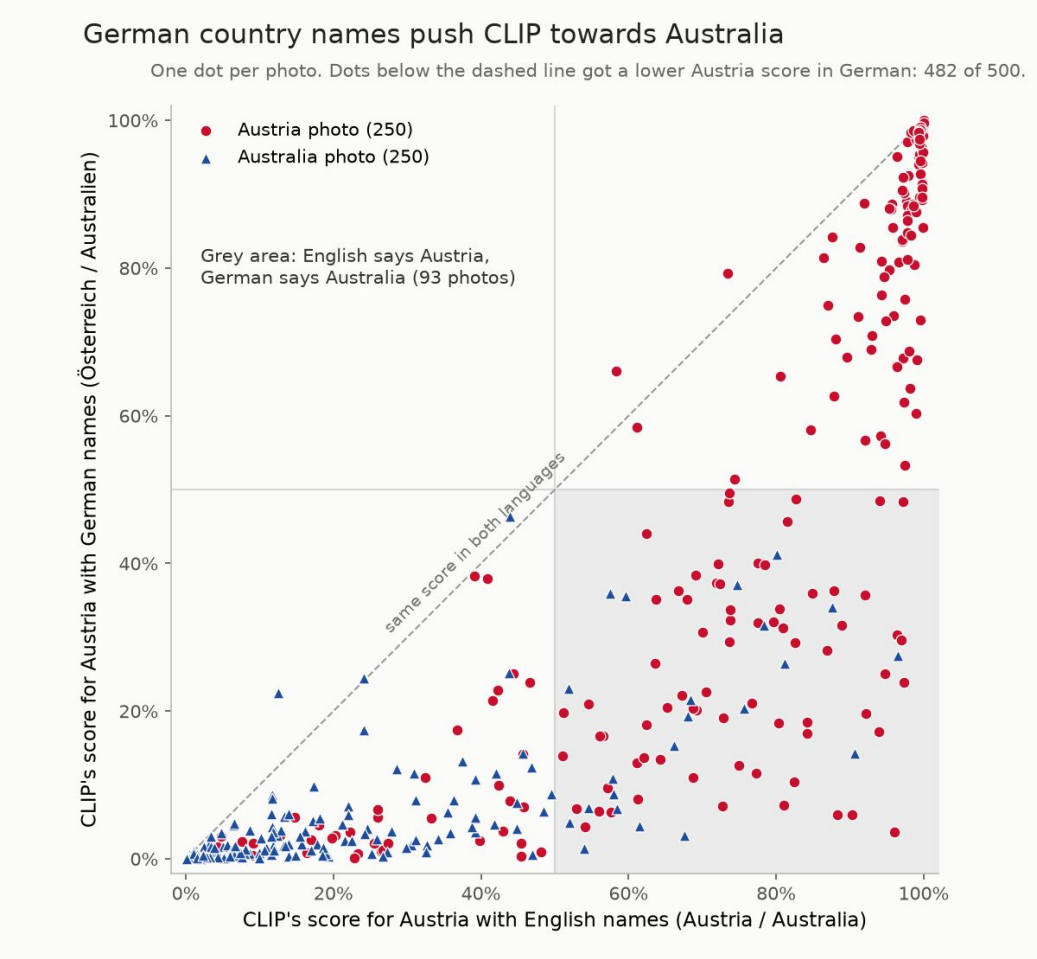
Text-side failure

In the worst cluster, concerts err 50% while image neighbours err 27%: the prompt wastes the signal.

All numbers from the notebook’s own printed outputs and its interpretation cell (cluster 8: 52 images → 2 sub-clusters, silhouette 0.06).

VISUAL 1 · PAIR 1

German names pull the scores towards Australia



Below the diagonal = German pulled towards Australia

482 of 500 dots sit below it. The effect is systematic, not noise.

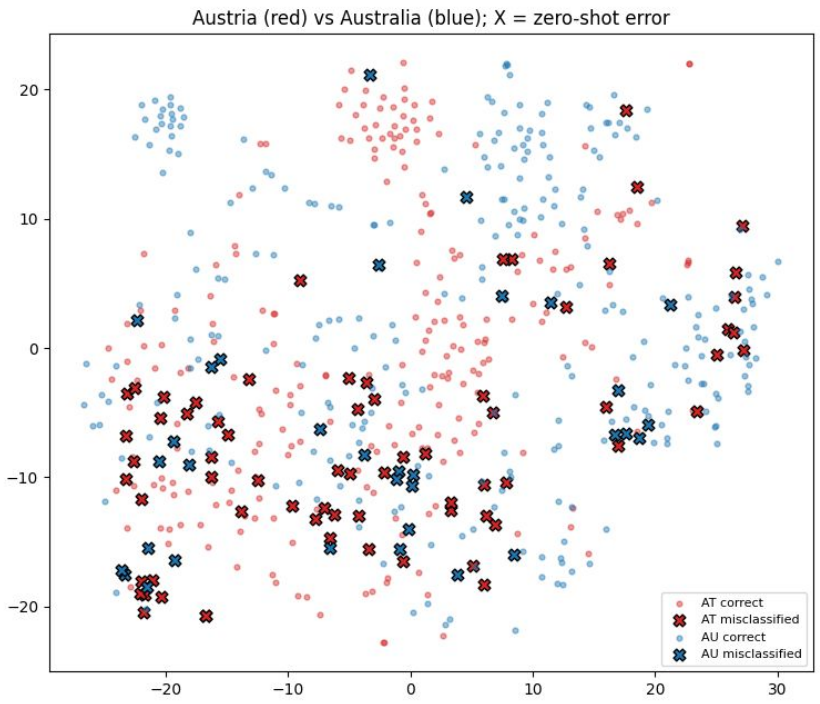
Grey box = 93 flips

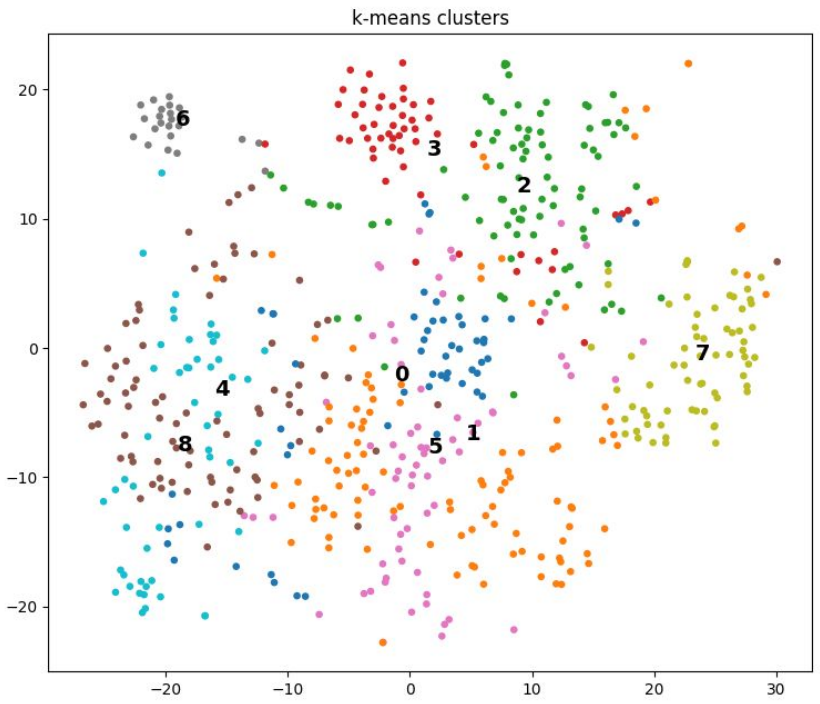
English said Austria, German said Australia: the words alone decided the country.

Pair 1 · language_shift_scatter.png

VISUAL 2 · PAIR 2

Errors pile up in the mixed centre





X = error

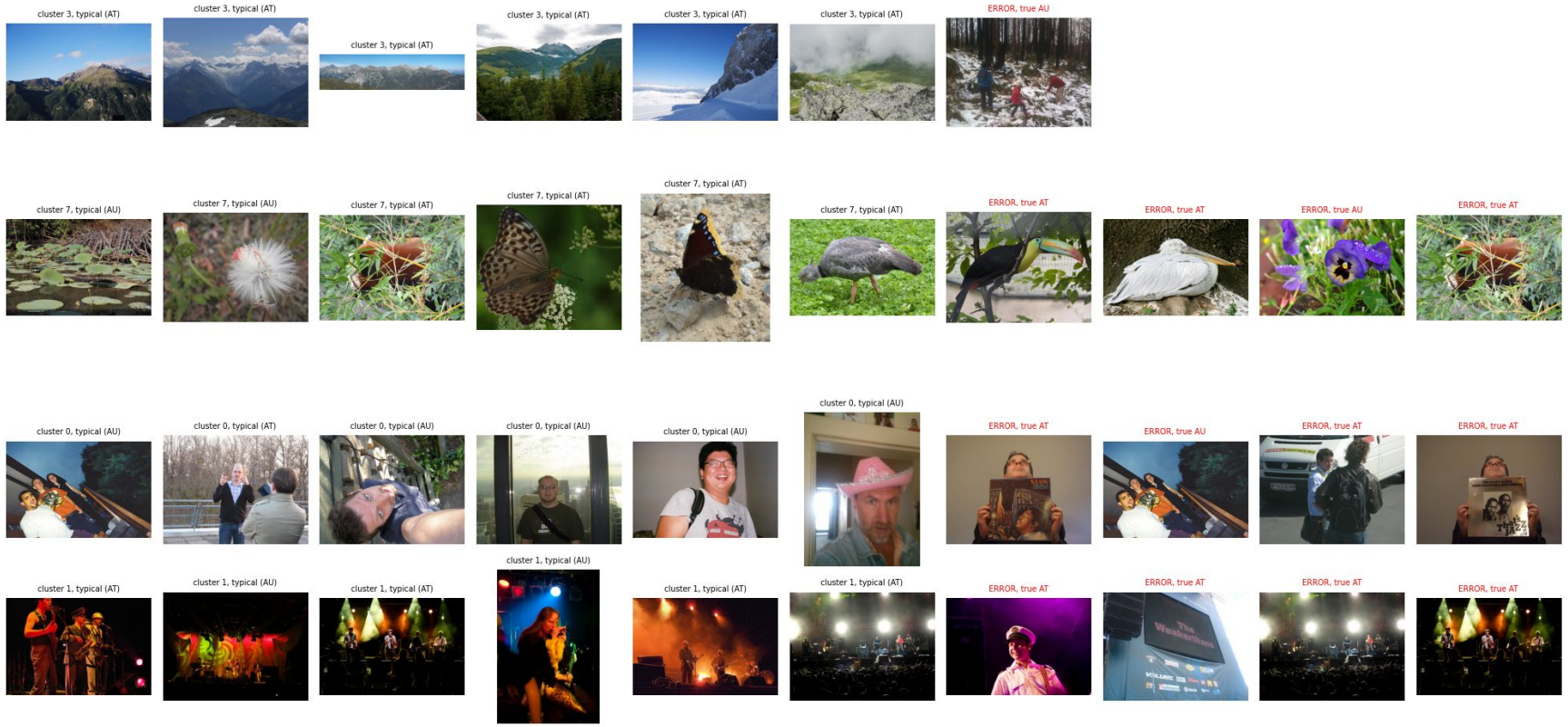
Errors cluster in the mixed centre, not at the country islands.

Colour = cluster

9 k-means clusters (silhouette 0.05); each is one kind of scene.

VISUAL 3 · PAIR 2

What the errors look like



Snow says Austria

Cluster 3's only error: an Australian snowy forest called Austria.

Wildlife says Australia

Cluster 7's worst errors: Austrian zoo animals, a toucan and a pelican.

One "bad" cluster

Two problems. Cluster 8: portraits 19% err vs concerts 50% (neighbours 27%): a text-side failure.

TWO SIDES, ONE STORY

CLIP reads scenes, not places

Text side · Pair 1

- 482 of 500 photos score lower for Austria under German names.
- 93 photos flip outright; the bias favours Australia.

Image side · Pair 2

- Place-specific scenes separate cleanly (≤ 7% error).
- Otherwise: scene stereotypes, biased towards Australia.

Convergent conclusion

Without place evidence, CLIP doesn’t guess at random. It falls back on postcard stereotypes, and the answer becomes unstable when you reword the prompt.

What GenAI did and didn’t do

Accepted

ChatGPT: project idea, CLIP + Country211; chose Language Shift and Prediction Explorer.

Claude Code (3 agents): 8 robustness fixes proposed for the Explorer: cache path, offline class names, Tk crash, label overlaps, stale predictions…

Rejected / open

uv setup rejected: “use his setup” (pip + PyCharm).

Drop .eval() rejected: explicit is better.

Re-encode JPEGs: still open for the pair.

Black-box warning (from our reflection)

“Easy to get results … it felt like a black box.” Fast ≠ understood: verification stayed human.

Where we stand

Phase I result

CLIP separates Austria from Australia **when the photo shows a place**. When it doesn't, it falls back on **scene stereotypes**, and the answer flips with the wording of the prompt.

Next steps

- Add the train split as a second dataset (≈ 300 MB more).
- Click a Language Shift dot to open that photo in the Explorer.
- Housekeeping: requirements.txt, README student IDs, video link.

Submission · 11 Nov 2026, 23:59

Done	Public repo: github.com/cristianknebel/AI-and-Visualization
Done	AI interaction record (ai-interactions/)
Done	GenAI reflection (ai_reflection_midpoint.md)
To do	Progress-update video, ≤ 5:00, unlisted link